Multiparty NOF search lower bounds for bit pigeonhole and semantic proof consequences

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Editorial scope. Counted once: full bit-PHP multiparty NOF search and semantic consequences. All load-bearing author reasons, probability argument, symmetry/bijection proofs, induction cases, reductions and semantic substitution continuations retained in editable TeX. Non-load-bearing background connective prose may be condensed. Original printed conventions, including Definition23/Claim27 indices and Section6.3 Q polynomial, retained without mathematical repair; formal PDF is the original source.

Allowed standard prerequisites (editorial).

- Randomized communication complexity, unique/set disjointness lower bounds and biased random-walk hitting time. Formal Section2 Theorems7–8 and AppendixA Theorem45, with original full33-entry bibliography. Standard imported results retained as cited prerequisites, not recursively reproved.

2 Preliminaries

Notations. We use the following notation: $[n] = \{1, 2, \ldots, n\}$. Let $S^{n\times m}$ denote the set of matrices of size $n \times m$ with elements from S. We denote by $0_{n\times m}$ the zero matrix of size $n \times m$ and by $1_{n\times m}$ the matrix of the same size containing only ones. For square matrices $A_1, \ldots, A_k$ we denote a diagonal block matrix with blocks $A_1, \ldots, A_k$ by $\mathrm{diag}(A_1, \ldots, A_k)$. For $x \in \{0, 1, \ldots, 2^k-1\}$ we denote a vector $(a_0, \ldots, a_{k-1}) \in \{0,1\}^k$ such that $x = \sum_{i=0}^{k-1} a_i 2^i$ by $\mathrm{bin}_k(x)$, i.e. $(a_0, \ldots, a_{k-1})$ is the reversed binary representation of x. For vectors $v_1, \ldots, v_n$ from a vector space over a field $\mathbb{F}$ we denote their linear span by $\mathrm{Span}(v_1, \ldots, v_n)$. We use coordinate-wise comparison of strings from $\{0,1\}^n$, i.e. for $x, y \in \{0,1\}^n$ we write $x \le y$ iff $x_i \le y_i$ for each $i \in [n]$. We denote the set of variables of a CNF-formula ϕ by $\mathrm{Vars}(\phi)$.

Communication complexity. We briefly recall some notions of communication complexity. For formal definition and details we refer to [24]. In the classic two-party randomized communication protocol with public randomness, Alice and Bob cooperate to compute a relation $Q \subseteq X \times Y \times Z$: Alice has an input $x \in X$ and Bob has an input $y \in Y$, their goal is to compute $z \in Z$ such that $(x, y, z) \in Q$. We assume that Alice and Bob have access to an arbitrary large random string of bits that is common for Alice and Bob. Let for every

$x \in X$ and $y \in Y$, $R^{\delta}_{\text{pub}}(Q, x, y)$ denote the minimal number of bits Alice and Bob need to transmit between each other so they both find a $z \in Z$ such that $(x, y, z) \in Q$ with probability at least $1 - \delta$ taken over the values of the common random string. And $R^{\delta}_{\text{pub}}(Q) := \max_{x \in X, y \in Y} R^{\delta}_{\text{pub}}(Q, x, y)$. We also consider multiparty communication protocols in the number on forehead (NOF) model that extends two-party protocols for an arbitrary number of parties. In this setting k parties cooperate to compute a relation $Q \subseteq X_1 \times X_2 \times \cdots \times X_k \times Y$. The ith party has $x_i \in X_i$ written on their forehead so they know all x_j for $j \neq i$, their goal is to compute $y \in Y$ such that $(x_1, x_2, \ldots, x_k, y) \in Q$. The parties communicate by taking turns broadcasting messages to all other parties until all parties learn the value of $y \in Y$ such that $(x_1, \ldots, x_k, y) \in Q$. In this model we also assume that all parties have access to a common random string of bits. Let $R^{\delta}_{\text{pub}}(Q, x_1, \ldots, x_k)$ for $x_1 \in X_1, \ldots, x_k \in X_k$ denote the minimal total number of bits transmitted until each party learns $y \in Y$ such that $(x_1, \ldots, x_k, y) \in Q$ with probability at least $1 - \delta$ taken over the set of values of the random string of bits. Also, let $R^{\delta}_{\text{pub}}(Q) := \max_{x_1 \in X_1, \ldots, x_k \in X_k} R^{\delta}_{\text{pub}}(Q, x_1, \ldots, x_k)$. Let f be a function from $X_1 \times X_2 \times \cdots \times X_k \to Y$. Then $R^{\delta}_{\text{pub}}(f)$ denotes $R^{\delta}_{\text{pub}}(Q_f)$, where $Q_f = \{(x_1, x_2, \ldots, x_k, y) \mid f(x_1, \ldots, x_k) = y\}$. We prove communication complexity lower bounds by reduction from different versions of the set disjointness problem. $\text{DISJ}_{k,n}$ is a function $\{0,1\}^{kn} \to \{0,1\}$ such that for every $x_1, \ldots, x_k \in \{0,1\}^n$ the following holds:

$$\text{DISJ}_{k,n}(x_1, \ldots, x_k) = \bigwedge_{j=1}^{n} \neg \left(\bigwedge_{i=1}^{k} (x_i)_j \right).$$

Let us define the communication promise problem $\text{UDISJ}_{k,n}$ in the k-party NOF model. For each $i \in [k]$ the string x_i is written on the forehead of the ith party, it is guaranteed that there exists at most one index $j \in [n]$ such that for every $i \in [k]$, $(x_i)_j = 1$. The goal is to compute $\text{DISJ}_{k,n}(x_1, \ldots, x_k)$.

Theorem7 ([31,32]). $R^{1/3}_{\text{pub}}(\text{UDISJ}_{k,n}) = \Omega(\sqrt{n}/(2^k k))$.

For $k = 2$ we omit the first index: $\text{DISJ}_n = \text{DISJ}_{2,n}$; in this case Theorem7 may be improved.

Theorem8 ([16]). $R^{1/3}_{\text{pub}}(\text{DISJ}_n) \geq R^{1/3}_{\text{pub}}(\text{UDISJ}_{2,n}) = \Omega(n)$.

Proof complexity. We consider refutational proof systems for the language of unsatisfiable CNF-formulas UNSAT. A refutation of $\phi \in \text{UNSAT}$ in a proof system Π is a sequence of Boolean functions (proof lines) such that each proof line either represents a clause of ϕ or derived from previous proof lines in the sequence via some sound inference rules. The last line of the proof is identically zero function. A proof system Π is defined by a representation of proof lines and by a set of admissible inference rules. It is required that the inference rules are polynomially verifiable i.e. there exists an algorithm that checks whether it is legitimate to derive a line L_0 from the lines $L_1, \ldots, L_k$. For example, in the Resolution proof lines are represented by clauses and the only inference rule is the resolution rule that allows deriving a clause $A \vee B$ from the clauses $A \vee x$ and $A \vee \neg x$. The size of a proof is the total size of all representations of proof lines in the proof. The length of a proof is the number of proof lines

in it. A tree-like proof is such a proof that every its line can be used as a premise of a rule at most once. For each proof system, we can also consider its tree-like version where all proofs are constrained to be tree-like. We also consider semantic refutational proof systems, where we drop the requirement for polynomial verification of inference rules i.e. we allow to derive any sound consequence from the premises. For such systems it is crucial to bound fan-in i.e. the number of the premises from which each proof line can be derived, otherwise, it would be possible to derive a contradiction from the clauses of the initial formula immediately. For example, it is well-known that Resolution is polynomially equivalent to a semantic proof system with fan-in 2 operating with clauses. A lower bound on the proof size in a semantic proof system implies a lower bound on the proof size in its syntactic counterpart because a syntactic proof is always a semantic proof that operates with the same class of proof lines. We define semantic Res($\oplus$) as a semantic proof system with fan-in 2 that operates with linear clauses. A linear clause is a disjunction of linear equations over $\mathbb{F}_2$: $\bigvee_{i=1}^{k}(f_i = a_i)$, where f_i is a linear form over $\mathbb{F}_2$ and $a_i \in \mathbb{F}_2$. Notice that an ordinary clause $\bigvee_{i\in P} x_i \vee \bigvee_{j\in N} \neg x_j$ can be represented by the linear clause $\bigvee_{i\in P}(x_i = 1) \vee \bigvee_{j\in N}(x_j = 0)$. For definition of syntactic version of Res($\oplus$) we refer to [15]; it is also proved there that syntactic and semantic Res($\oplus$) are polynomially equivalent. We define semantic Res(PC_d) as a semantic proof system with fan-in 2 that operates with disjunctions of equations of type $f = 0$, where f is a degree-d polynomial over $\mathbb{F}_2$. Notice that semantic Res(PC_1) is exactly semantic Res($\oplus$). For the definition of the syntactic version of Res(PC_d) we refer to [19]. Following [1] we define Th(k) as a semantic proof system with fan-in 2 that operates with polynomial inequalities $g \geq 0$, where g is a polynomial of degree at most k with integer coefficients and Boolean variables. A clause $\bigvee_{i\in P} x_i \vee \bigvee_{j\in N} \neg x_j$ can be represented by an inequality $\sum_{i\in P} x_i + \sum_{j\in N}(1 - x_j) - 1 \geq 0$.

Proof complexity and communication complexity. For an unsatisfiable CNF-formula ϕ we define the communication problem Search(ϕ). Search(ϕ) is the following problem: given an assignment of the variables of the unsatisfiable CNF ϕ, find a clause that is falsified by this assignment. It is assumed that variables of ϕ are somehow partitioned between the parties. Following the paper [9] we consider a semantic proof system $\mathrm{T}^{\mathrm{cc}}(k, c)$ that models many interesting syntactic and semantic proof systems. The proof lines in $\mathrm{T}^{\mathrm{cc}}(k, c)$ can be arbitrary Boolean functions having the following property: for every proof line C and every partition of variables of C between k parties, the NOF k-party randomized communication complexity of C is at most c w.r.t. this partition. We also define a semantic proof system $\mathrm{T}^{\mathrm{cc}}_{\mathrm{os}}(k, c)$ that is a subsystem of $\mathrm{T}^{\mathrm{cc}}(k, c)$ with the restriction that a communication protocol for proof lines must have a one-sided error: if the value of a proof line is zero, then the protocol should return zero with probability 1. For example, $\mathrm{T}^{\mathrm{cc}}(2, 2)$ simulates Resolution; $\mathrm{T}^{\mathrm{cc}}(2, O(1))$ simulates Res($\oplus$) [22,15]; $\mathrm{T}^{\mathrm{cc}}(k, O(k^3 \log^2 n))$, where n is the number of variables in a refuted formula, simulates Th($k-1$) [9]. In Section3 we show that $\mathrm{T}^{\mathrm{cc}}_{\mathrm{os}}(d + 1, O(1))$ simulates Res(PC_d). The following connection between the communication complexity of Search(ϕ) and tree-like proof complexity of ϕ is known.

Lemma9 ([1,9]). If a CNF formula ϕ has a tree-like $\mathrm{T}^{\mathrm{cc}}(k, c)$ refutation of

length ℓ then, over any k-partition of the variables, there is a randomized bounded-error k-party NOF protocol for Search(ϕ) with communication cost $O(c \cdot \log \ell \log\log \ell)$.

In Section3 we show that for $\mathrm{T}^{\mathrm{cc}}_{\mathrm{os}}(k,c)$ the bound can be improved, see Remark14.

Basic formulas. A CNF formula PHP^m_n encodes the pigeonhole principle; PHP^m_n states that it is possible to put m pigeons into n holes such that every pigeon flies to at least one hole and at most one pigeon flies to each hole. PHP^m_n depends on variables $p_{i,j}$ for $i \in [m]$ and $j \in [n]$ and $p_{i,j} = 1$ iff the i-th pigeon flies to the j-th hole. PHP^m_n is the conjunction of $m(m-1)n/2$ hole axioms and m pigeons axioms. For every $i \in [m]$ PHP^m_n contains a pigeon axiom $(p_{i,1} \vee p_{i,1} \vee \cdots \vee p_{i,n})$. And for every $j \in [n]$ and every $k \neq \ell \in [n]$, PHP^m_n contains a hole axiom $(\neg p_{k,j} \vee \neg p_{\ell,j})$. PHP^m_n is unsatisfiable iff $m > n$. For an undirected graph $G(V,E)$, the formula PM_G encodes in CNF that G has a perfect matching. The formula PM_G has $|E|$ variables, each of them corresponds to an edge of G, x_e is the variable corresponding to $e \in E$.

$$\mathrm{PM}_G = \bigwedge_{v \in V} \left(\left(\bigvee_{e \text{ is incident to } v} x_e \right) \wedge \bigwedge_{e_1 \neq e_2 \text{ are incident to } v} (\neg x_{e_1} \vee \neg x_{e_2}) \right).$$

PM_G is unsatisfiable iff G does not have a perfect matching.

Theorem10 ([26]). Let G be a graph with n vertices, which has no perfect matching. Then the formula PM_G has a tree-like Res($\oplus$) refutation of size $2^{O(n)}$.

Proposition11 ([14]). Let G be a graph with an odd number of vertices. Then the formula PM_G has a tree-like Res($\oplus$) refutation of size poly(n).

The binary pigeonhole principle $\mathrm{BPHP}^m_{2^\ell}$ states that there are m different ℓ-bit binary strings $s_1, s_2, \ldots, s_m$. $\mathrm{BPHP}^m_{2^\ell}$ has $m\ell$ variables corresponding to the bits of s_i for $i \in [m]$. Then $\mathrm{BPHP}^m_{2^\ell} = \bigwedge_{i \neq j \in [m]} s_i \neq s_j$, where the predicate $s_i \neq s_j$ is encoded as a 2ℓ-CNF formula of size 2^ℓ as follows: $\bigwedge_{\alpha \in \{0,1\}^\ell} (s_i \neq \alpha \vee s_j \neq \alpha)$; notice that the predicate $(s_i \neq \alpha \vee s_j \neq \alpha)$ can be represented by a clause with 2ℓ literals. If $m > 2^\ell$, then the formula $\mathrm{BPHP}^m_{2^\ell}$ is unsatisfiable. Let ϕ be a CNF formula with n variables, and $g : \{0,1\}^k \to \{0,1\}$ be a Boolean function. Then $\phi \circ g$ denotes a CNF formula on kn variables that represents $\phi(g(\vec{x}_1), g(\vec{x}_2), \ldots, g(\vec{x}_n))$, where $\vec{x}_i$ denotes a vector of k new variables. $\phi \circ g$ is constructed by applying the substitution to every clause C of ϕ and converting the resulting function $C \circ g$ to CNF in some fixed way.

Original introduction: proof-complexity consequences and scope A lower bound on the tree-like communication complexity of Search(ϕ) implies a lower bound on the size of tree-like refutations in the corresponding proof system, and on the size of dag-like refutations using small space (size-space tradeoffs [9,12]). The usual translation from tree-like refutations of size S, or small-space realizations of dag-like refutations of size S, gives communication cost $O(\log S \log\log S \cdot c)$, with c the communication cost of a proof line; sometimes it improves to $O(\log S \cdot c)$. Randomized two-party protocols correspond

to Res($\oplus$) [15], whose lines are disjunctions of linear equations over $\mathbb{F}_2$. Real k-party NOF protocols correspond to semantic Th$(k-1)$ [1], whose lines are integer-coefficient polynomial inequalities of degree at most $k-1$ on Boolean variables. Randomized k-party NOF protocols correspond to semantic $\mathrm{T}^{\mathrm{cc}}(k,c)$, whose lines are arbitrary predicates of randomized NOF communication cost at most c. For small c this simulates Th$(k-1)$ and Res(PC$_{k-1}$); the latter's lines are disjunctions of polynomial equations over $\mathbb{F}_2$ of bounded degree. Theorem4 yields the lower bound $\exp(2^{\Omega(n/k)})$ on the size of tree-like $\mathrm{T}^{\mathrm{cc}}(k,c)$ (and Th$(k-1)$) refutations of $\mathrm{BPHP}_{2^n}^{2^n+2^{n+k-\lfloor n/k \rfloor}}$. Hrubes and Pudlak [10] obtained lower bounds for dag-like two-party real protocols with the same variable partition and arbitrary $m > 2^\ell$; the results are formally incomparable because the present randomized model is stronger and multiparty, while [10] covers dag-like protocols and arbitrary weak binary PHP. The randomized communication lower bound is nontrivial for $k \le \log^{1-\varepsilon} n$ with $\varepsilon > 0$, yielding superpolynomial tree-like Th(k) lower bounds for BPHP_n^m. Proposition6 gives short tree-like Th$(\log n)$ refutations. Together with [10] it implies tree-like Th$(\log n)$ cannot be simulated by dag-like Th(1). Theorem4 and Proposition6 imply that $\mathrm{BPHP}_{2^\ell}^{2^\ell+1}$ separates tree-like Th$(\log n)$ from tree-like Th(k) for $k \le \log^{1-\varepsilon} n$. Here $n = (2^\ell+1)\ell$ is the number of variables; Proposition6 gives polynomial size, while Theorem4 gives at least $\exp(\exp(\Omega(\log^\varepsilon n)))$, which is superpolynomial. The abstract states rank lower bounds among the general consequences of randomized NOF search bounds; these are contextual consequences rather than a separately counted claim.

3 Communication protocols from tree-like Res(PCd) proofs

Let ϕ be an unsatisfiable CNF formula with n variables. Let us define the communication problem $\oplus_k \mathrm{Search}(\phi)$ with k parties as follows. Assume that the ith party has an assignment $\alpha_i \in \{0,1\}^n$ written on the forehead. They aim to find a clause of ϕ falsified by the assignment $\sum_{i=1}^k \alpha_i$ (all sums of boolean vectors are computed modulo 2).

Lemma1. Let ϕ be an unsatisfiable CNF formula. If there exists a tree-like Res(PC$_d$) proof of ϕ of length m, then $R^{1/3}_{\mathrm{pub}}(\oplus_{d+1}\mathrm{Search}(\phi)) = O(d \cdot \log m)$.

A slightly weaker version of the following lemma was implicitly proved in [15]:

Lemma12 (see proof of Theorem3.11 from [15]). Let T be a binary tree with m vertices such that the ith vertex is labeled with $a_i \in \{0,1\}$ with the hereditary property: for each inner vertex i with direct descendants c_1 and c_2, if $a_i = 1$, then $a_{c_1} = 1$ or $a_{c_2} = 1$. We also assume that if r is the root of T, then $a_r = 1$. Assume that we have a one-sided bounded error oracle access to a_i i.e. if we request a value of a_i and $a_i = 0$ we get 1 with probability at most 1/2 and0 with probability at least 1/2; if $a_i = 1$ we get 1 with probability 1. Then there exists an algorithm A that with probability at least 2/3 returns a leaf ℓ of T with $a_\ell = 1$ and makes $O(\log m)$ oracle queries to $a_1, \ldots, a_m$.

Proof. See AppendixA.

Proof of Lemma1. Let $F_1, \ldots, F_m$ be a tree-like Res(PC$_d$)-refutation of ϕ with the underlying tree T, where vertices of T are identified with $[m]$. Then the leaves of T correspond to the clauses of ϕ and m is the root of T. Let $\alpha_1, \ldots, \alpha_{d+1}$ be the assignments written on the foreheads of $d+1$ parties. Let $\alpha = \sum_{i=1}^{d+1} \alpha_i$. Let $a_i = 1$ iff α falsifies F_i for $i \in [m]$. Then $a_m = 1$ since F_m is identically false. For any inner node v of T, if $a_v = 1$ then for the direct descendants of v, c_1 and c_2 either $a_{c_1} = 1$ or $a_{c_2} = 1$. In the next paragraphs we show that for any $i \in [m]$ there exists a NOF $(d+1)$-party protocol that computes a_i given that for each $j \in [d+1]$ the jth party has α_j written on their forehead such that

- the protocol transmits $O(d)$ bits;
- the protocol has one-sided bounded error: if $a_i = 1$ then the protocol returns 1 with probability 1 and if $a_i = 0$ the protocol returns 0 with probability at least 1/2.

Then we use this protocol to compute a_i as an oracle in the algorithm given by Lemma12 and thus show that there is a NOF $(d+1)$-party protocol computing $\oplus_{d+1}\mathrm{Search}(\phi)$ with communication cost $O(d \log m)$. Now we show that for every $\ell \in [m]$, $F_\ell(\alpha)$ can be computed by a $(d+1)$-party NOF protocol with one-sided error using $O(d)$ bits of communication. Let $F_\ell = \bigvee_{j=1}^{t}(f_j = 1)$, where $f_1, \ldots, f_t$ are polynomials over $\mathbb{F}_2$ of degree at most d. Let $z_1, \ldots, z_n$ be the variables of ϕ. Let us introduce new variables $y_{1,1}, \ldots, y_{1,n}, \ldots, y_{d+1,1}, \ldots, y_{d+1,n}$ and assume that for each $i \in [d+1]$ the ith party has the value of variables $y_{i,1}, y_{i,2}, \ldots, y_{i,n}$ written on the forehead or in other words α_i assigns values of $y_{i,1}, y_{i,2}, \ldots, y_{i,n}$. Let $\bar{f}_j$ denote f_j after substitution $z_\ell := y_{1,\ell} + y_{2,\ell} + \cdots + y_{d+1,\ell}$ for $\ell \in [n]$; $j \in [t]$. Since for all $j \in [t]$, $\deg \bar{f}_j = \deg f_j \leq d$, we can represent $\bar{f}_j = \bar{f}_j^{(1)} + \cdots + \bar{f}_j^{(d+1)}$ such that $\bar{f}_j^{(s)}$ does not contain variables $y_{s,1}, \ldots, y_{s,n}$ for each $s \in [d+1]$. Then the ith party can compute $\bar{f}_1^{(i)}(\alpha_1, \ldots, \alpha_{d+1}), \ldots, \bar{f}_t^{(i)}(\alpha_1, \ldots, \alpha_{d+1})$. Notice that $F_\ell = \neg(\bigwedge_{j=1}^{t}(f_j = 0))$. The final step of the protocol exploits the idea used to construct a short randomized communication protocol for equality. Take a random uniformly distributed vector $(e_1, \ldots, e_t) \in \mathbb{F}_2^t$. Then all parties compute

$$\sum_{j=1}^{t} e_j f_j(\alpha) = \sum_{i=1}^{d+1} \underbrace{\sum_{j=1}^{t} e_j \bar{f}_j^{(i)}}_{i\text{th party}}$$

with $O(d)$ bits of communication and the protocol halts. To bound the error probability we use the following well-known statement:

Proposition13 (Random subsum principle). For any $x \in \mathbb{F}_2^k \setminus \{0^k\}$,

$$\Pr_{y \leftarrow U(\mathbb{F}_2^k)}\left[\sum_{i=1}^{k} y_i x_i = 1\right] = \frac{1}{2}.$$

If $F_\ell(\alpha) = 1$ then $\Pr[\sum_{j=1}^{t} e_j f_j(\alpha) \neq 0] = 1/2$ by the random subsum principle. If $F_\ell(\alpha) = 0$, then $\Pr[\sum_{j=1}^{t} e_j f_j(\alpha) = 0] = 1$. □

Remark14. Similarly to the proof of Lemma1 one can prove that if an unsatisfiable CNF formula ϕ has a tree-like $\mathrm{T}^{\mathrm{cc}}_{\mathrm{os}}(k,c)$ refutation of length ℓ, then for any k-partition of the variables, there is a randomized bounded-error k-party NOF protocol for Search(ϕ) with communication cost $O(c\log\ell)$. Thus, the bound from Lemma9 can be slightly improved in the case of one-sided error.

Theorem 3. Let ℓ and k be natural numbers such that $2 \le k \le \ell - 7$. Then

$$R^{1/3}_{\mathrm{pub}}(\oplus_k \mathrm{Search}(\mathrm{BPHP}^{2^\ell+2^k}_{2^\ell})) = \Omega\left(\frac{2^{\ell/2}}{k2^{3k/2}}\right).$$

For $k=2$ the stronger bound holds: $R^{1/3}_{\mathrm{pub}}(\oplus_2 \mathrm{Search}(\mathrm{BPHP}^{2^\ell+4}_{2^\ell})) = \Omega(2^\ell)$.

We consider a combinatorial analogue of the communication problem $\oplus_k \mathrm{Search}(\mathrm{BPHP}^m_{2^\ell})$. Assume that each of k parties gets m binary strings from $\{0,1\}^\ell$, where $m > 2^\ell$. The ith party has numbers $a_{i,1},\dots,a_{i,m} \in \{0,1\}^\ell$ on their forehead. Based on these strings we form m vectors from $\mathbb{F}_2^\ell$: $x_1,\dots,x_m$, where $x_j = \sum_{i=1}^k a_{i,j}$. The goal is to find different $t,s \in [m]$ such that $x_t = x_s$. We denote this problem by $\oplus_k \mathrm{BPHP}^m_{2^\ell}$. It is straightforward that $R^{1/3}_{\mathrm{pub}}(\oplus_k \mathrm{Search}(\mathrm{BPHP}^m_{2^\ell})) \ge R^{1/3}_{\mathrm{pub}}(\oplus_k \mathrm{BPHP}^m_{2^\ell})$, hence it is sufficient to prove a lower bound on the latter.

Theorem 18. Let ℓ, k be natural numbers such that $2 \le k \le \ell - 7$. Then

$$R^{1/3}_{\mathrm{pub}}(\oplus_k \mathrm{BPHP}^{2^\ell+2^k}_{2^\ell}) = \Omega\big(R^{1/3}_{\mathrm{pub}}(\mathrm{UDISJ}_{k,2^{\ell-k}-1}) - \ell\big).$$

Corollary 19. $R^{1/3}_{\mathrm{pub}}(\oplus_k \mathrm{BPHP}^{2^\ell+2^k}_{2^\ell}) = \Omega(2^{\ell/2}/(k2^{3k/2}))$. For $k=2$: $R^{1/3}_{\mathrm{pub}}(\oplus_2 \mathrm{BPHP}^{2^\ell+4}_{2^\ell}) = \Omega(2^\ell)$.

Proof of Corollary 19. Follows from Theorem 18 and Theorem 7; for $k=2$ apply Theorem 8. □

Theorem 3 immediately follows from Corollary 19.

5.1 Warm-up example

Consider $\mathrm{Distinct}_{k,\ell}$: each of k parties has a matrix from $\mathbb{F}_2^{2^\ell \times \ell}$ on their forehead. The goal is to determine whether all rows of the sum of these matrices are distinct. A version without the xor-gadget is referred to as Element Distinctness (ED) [25].

Proposition 20. $R^{1/3}_{\mathrm{pub}}(\mathrm{Distinct}_{k,\ell}) \ge R^{1/3}_{\mathrm{pub}}(\mathrm{UDISJ}_{k,2^{\ell-k}})$.

Let S_k denote the set of matrices from $\{0,1\}^{2^k\times k}$ with all distinct rows. Let $K_k \in \{0,1\}^{2^k\times k}$ have ith row $\mathrm{bin}_k(i-1-((i-1) \bmod 2))$, i.e. rows $\mathrm{bin}_k(0), \mathrm{bin}_k(0), \mathrm{bin}_k(2), \mathrm{bin}_k(2), \dots, \mathrm{bin}_k(2^{k-1}-2), \mathrm{bin}_k(2^{k-1}-2)$. Every row starts with zero and appears exactly twice. The following combinatorial lemma is proved in Subsection 5.4.

Lemma 21. There exist matrices $A_1(0), A_1(1), \dots, A_k(0), A_k(1) \in \mathbb{F}_2^{2^k\times k}$ such that $\sum_{i=1}^k A_i(1) = K_k$ and for all $b_1,\dots,b_k \in \{0,1\}$, if $\bigwedge_{i=1}^k b_i = 0$, then $\sum_{i=1}^k A_i(b_i) \in S_k$.

Proof of Proposition 20. Let $(x_{i,1}, \ldots, x_{i,2^{\ell-k}})$ be an input of the ith party of the problem $\mathrm{UDISJ}_{k,2^{\ell-k}}$. For all $i \in [k]$ we construct a matrix D_i of size $2^\ell \times \ell$ and put it on the forehead of the ith party. Let $A_i(b)$ for $i \in [k]$, $b \in \{0,1\}$ be matrices of size $2^k \times k$ from Lemma21. Let J_t for $t \in [1, \ldots, 2^{\ell-k}]$ be a matrix of size $2^k \times (\ell - k)$ such that all its rows are equal to $\mathrm{bin}_{\ell-k}(t-1)$. Let us define

$$D_1 := \begin{pmatrix} J_1 & A_1(x_{1,1}) \\ \vdots & \vdots \\ J_j & A_1(x_{1,j}) \\ \vdots & \vdots \\ J_{2^{\ell-k}} & A_1(x_{1,2^{\ell-k}}) \end{pmatrix}; \quad D_i := \begin{pmatrix} 0_{2^k \times (\ell-k)} & A_i(x_{i,1}) \\ \vdots & \vdots \\ 0_{2^k \times (\ell-k)} & A_i(x_{i,j}) \\ \vdots & \vdots \\ 0_{2^k \times (\ell-k)} & A_i(x_{i,2^{\ell-k}}) \end{pmatrix} \quad (i \in \{2, \ldots, k\}).$$

By Lemma21, the matrix $D_1 + D_2 + \cdots + D_k$ has the following property: for all $j \in [2^{\ell-k}]$, its submatrix formed by the rows with numbers from $[2^k \cdot (j-1) + 1, 2^k \cdot j]$ has two equal rows if and only if $x_{1,j} = x_{2,j} = \cdots = x_{k,j} = 1$. Thus, the communication complexity of $\mathrm{UDISJ}_{k,2^{\ell-k}}$ is at most the communication complexity of $\mathrm{Distinct}_{k,\ell}$. □

5.2 Proof of Theorem 18

We modify Proposition20 to reduce $\mathrm{UDISJ}_{k,2^{\ell-k}-1}$ to $\oplus_k \mathrm{BPHP}^{2^\ell+2^k}_{2^\ell}$ by adding fake rows not corresponding to the disjointness input and randomizing to hide them among other rows.

Proof of Theorem 18. For $N > 2^\ell$, define $\mathrm{ROW} \oplus_k \mathrm{BPHP}^N_{2^\ell}$: party i has $M_i \in \mathbb{F}_2^{N \times \ell}$ on their forehead and the goal is to find the value of a row appearing at least twice in $M_1 + \cdots + M_k$, rather than the indices of equal rows.

Claim 22. If $R^{1/3}(\oplus_k \mathrm{BPHP}^N_{2^\ell}) \le t$, there exists a protocol P for $\mathrm{ROW} \oplus_k \mathrm{BPHP}^N_{2^\ell}$ using $O(t+\ell)$ bits that returns a correct answer or $\perp$, with $\Pr[P(M_1, \ldots, M_k) = \perp] \le 1/100$ for every input.

Proof. P executes a randomized protocol for $\oplus_k \mathrm{BPHP}^N_{2^\ell}$ and verifies its answer by transferring additional $O(\ell)$ bits. The probability of failure can be reduced by repetition. □

We describe a protocol for $\mathrm{UDISJ}_{k,2^{\ell-k}-1}$ using P for $\mathrm{ROW} \oplus_k \mathrm{BPHP}^{2^\ell+2^k}_{2^\ell}$. Inputs are $x_1, \ldots, x_k \in \{0,1\}^{2^{\ell-k}-1}$; $x_{i,j}$ is bit j of x_i and $\vec{x} = (x_1, \ldots, x_k)$.

Important matrices Let $\gamma : [2^{\ell-k}-1] \cup \{*\} \to \{0,1\}^{\ell-k}$ be bijective. Define matrices $D_1(x_1, \gamma), D_2(x_2), \ldots, D_k(x_k)$ of size $(2^\ell + 2^k) \times \ell$. Let $A_i(b)$ be from Lemma21; for every $t \in \{0,1\}^{\ell-k}$ let J_t have size $2^k \times (\ell - k)$ with every row t. Fix $W \in S_k$. Define

$$D_1(x_1, \gamma) := \begin{pmatrix} J_{\gamma(1)} & A_1(x_{1,1}) \\ \vdots & \vdots \\ J_{\gamma(j)} & A_1(x_{1,j}) \\ \vdots & \vdots \\ J_{\gamma(2^{\ell-k}-1)} & A_1(x_{1,2^{\ell-k}-1}) \\ J_{\gamma(*)} & W \\ J_{\gamma(*)} & W \end{pmatrix};$$

for $i \in [k] \setminus \{1\}$,

$$D_i(x_i) := \begin{pmatrix} 0_{2^k \times (\ell-k)} & A_i(x_{i,1}) \\ \vdots & \vdots \\ 0_{2^k \times (\ell-k)} & A_i(x_{i,j}) \\ \vdots & \vdots \\ 0_{2^k \times (\ell-k)} & A_i(x_{i,2^{\ell-k}-1}) \\ 0_{2^k \times (\ell-k)} & 0_{2^k \times k} \\ 0_{2^k \times (\ell-k)} & 0_{2^k \times k} \end{pmatrix}.$$

The last 2^{k+1} rows of $D_1(x_1, \gamma)$ contain each row exactly twice. Define $H_{\vec{x}}(\gamma) := D_1(x_1, \gamma) + D_2(x_2) + \cdots + D(x_k)$. Lemma21 gives the following property in the standard basis.

Definition 23. Let $M \in \mathbb{F}_2^{(2^k+2^\ell)\times \ell}$, $\gamma : [2^{\ell-k}-1] \cup \{*\} \to \{0,1\}^{\ell-k}$ bijective, and $e_1, \ldots, e_\ell$ a basis in $\mathbb{F}^\ell$. M satisfies the key property w.r.t. $(\gamma, \vec{x})$ in this basis if:

- For a row s among the last 2^{k+1} rows, the first $\ell - k$ basis coordinates are $\gamma(*)_1, \ldots, \gamma(*)_{\ell-k}$ and s appears exactly twice.
- For a row s with index in $[2^k(i-1)+1; 2^k i]$, $i \in [2^{\ell-k}-1]$, the first $\ell-k$ basis coordinates are $\gamma(i)_1, \ldots, \gamma(i)_{\ell-k}$. If $\bigwedge_{j=1}^k x_{i,j} = 0$ it appears exactly once; if $\bigwedge_{j=1}^k x_{i,j} = 1$ it appears exactly twice and its $(\ell-k+1)$th basis coordinate is 0.

Consider an invertible matrix $E \in \mathbb{F}_2^{\ell\times\ell}$. Let $e_1, e_2, \ldots, e_\ell$ be the rows of E. Since E is invertible, $e_1, e_2, \ldots, e_\ell$ form a basis. Let us define $C_{\vec{x}}(\gamma, E) := H(\vec{x}, \gamma)E$. Rows of $C_{\vec{x}}(\gamma, E)$ can be viewed as vectors with coordinates in the basis $e_1, e_2, \ldots, e_\ell$ corresponding to the rows of $H(\vec{x}, \gamma)$. Hence, $C_{\vec{x}}(\gamma, E)$ satisfies the key property w.r.t. $(\gamma, \vec{x})$ in the basis $(e_1, e_2, \ldots, e_\ell)$. Define $\mathrm{Fake}(\gamma, E) \subseteq \mathbb{F}_2^\ell$ to be the set of the last 2^{k+1} rows of $C_{\vec{x}}(\gamma, E)$. This set does not depend on $\vec{x}$. Its rows appear exactly twice.

Random variables Random variables are bold. Public randomness consists of:

- Uniform random bijection $\boldsymbol{\gamma} : [2^{\ell-k}-1] \cup \{*\} \to \{0,1\}^{\ell-k}$.
- Uniform random invertible matrix $\mathbf{E} \in \mathbb{F}_2^{\ell\times\ell}$.
- Uniform random permutation $\boldsymbol{\pi}$ of $[2^\ell + 2^k]$, with corresponding permutation matrix $M_{\boldsymbol{\pi}}$: $(M_{\boldsymbol{\pi}})_{i,j} = 1 \iff \boldsymbol{\pi}(i) = j$.
- Uniform random matrices $\boldsymbol{\Delta}_1, \ldots, \boldsymbol{\Delta}_k \in \mathbb{F}_2^{(2^\ell+2^k)\times\ell}$ subject to $\sum_i \boldsymbol{\Delta}_i = 0$.

Define $\mathbf{P}_i = M_{\boldsymbol{\pi}} D_i(x_i)\mathbf{E} + \boldsymbol{\Delta}_i$ for $i \geq 2$ and $\mathbf{P}_1 = M_{\boldsymbol{\pi}} D_1(x_1, \boldsymbol{\gamma})\mathbf{E} + \boldsymbol{\Delta}_1$. The addition of $\boldsymbol{\Delta}_i$ makes each $\mathbf{P}_i$ indistinguishable from a random matrix. Moreover, $\sum_i \mathbf{P}_i = M_{\boldsymbol{\pi}} C_{\vec{x}}(\boldsymbol{\gamma}, \mathbf{E})$, obtained by permuting its rows. Let N be a constant to be chosen later.

Algorithm 1: Protocol T solving $\mathrm{UDISJ}_{k,2^{\ell-k}-1}$**.** Input $x_1, \ldots, x_k \in \{0,1\}^{2^{\ell-k}-1}$; x_i is on party i's forehead.

$Z := \varnothing$
loop
repeat N times
  Sample $\pi \leftarrow \boldsymbol{\pi}$, $E \leftarrow \mathbf{E}$, $\gamma \leftarrow \boldsymbol{\gamma}$, $\vec{\Delta} \leftarrow \vec{\boldsymbol{\Delta}}$ ▷ Fresh public bits
  $P_1 := M_\pi D_1(x_1, \gamma)E + \Delta_1$
▷ Computable by parties $2, \ldots, k$.
  $P_i := M_\pi D_i(x_i)E + \Delta_i$ for $i \geq 2$
▷ Computable by all parties except i.
  $z := P(P_1, \ldots, P_k)$
▷ Fresh random bits for P; P_i is on party i's forehead.
  $Z := Z \cup \{z\}$
end loop
if $Z = \{\perp\}$ **then**
  return $\perp$
else if $Z \setminus \{\perp\} \subseteq \mathrm{Fake}(\gamma, E)$ **then**
  return 1
▷ Most likely there are no repeated rows except Fake; the key property gives $\mathrm{DISJ}(x_1, \ldots, x_k) = 1$.
end if
return 0

Protocol analysis Let us analyze the protocol T. Since it executes the protocol P a constant number of times, T transmits $O(t + \ell)$ bits. Assume that $x_1, x_2, \ldots, x_k$ is a 1-instance of $\mathrm{UDISJ}_{k,2^\ell+2^k}$. Then by the key property of $C_{\vec{x}}(\boldsymbol{\gamma}, \mathbf{E})$ all repeated rows of $\sum_{i=1}^k \mathbf{P}_i$ are in $\mathrm{Fake}(\boldsymbol{\gamma}, \mathbf{E})$, hence the protocol T returns either $\perp$ or the correct answer. Since P is executed N times independently, the probability that $Z = \{\perp\}$ is at most $1/100^N$, hence T returns1 with probability at least $1 - 1/100^N$. The rest of the proof is devoted to the analysis of the case, where $x_1, x_2, \ldots, x_k$ is a 0-instance of $\mathrm{UDISJ}_{k,2^\ell+2^k}$. This is the most technically involved part of the proof. So it is a good point to give a large scale overview of the further proof strategy. Our goal is to show that if $x_1, x_2, \ldots, x_k$ is a 0-instance of $\mathrm{UDISJ}_{k,2^\ell+2^k}$, then the probability that $P(\mathbf{P}_1, \ldots, \mathbf{P}_k)$ returns a value from $\mathrm{Fake}(\boldsymbol{\gamma}, \mathbf{E})$ is bounded by some constant less than1. The random variable $P(\mathbf{P}_1, \ldots, \mathbf{P}_k)$ depends on random bits used by the protocol P and on random bits needed for sampling $\mathbf{P}_1, \ldots, \mathbf{P}_k$. Let R denote the set of all random strings used by the protocol P (i.e. we assume that P sample a random string from R and use it as public randomness) and S denote the set of all random strings used for sampling $\mathbf{P}_1, \ldots, \mathbf{P}_k$. We would like to construct two bijections α and β on the set S such that for every $s \in S$ the following two properties hold.

1. The three values of random variable $(\mathbf{P}_1, \ldots, \mathbf{P}_k)$ sampled using three strings $s, \alpha(s)$ and $\beta(s)$ as a random source, are the same.

2. Let (γ, E), $(\gamma_\alpha, E_\alpha)$ and (γ_β, E_β) be values of the random variable $(\boldsymbol{\gamma}, \mathbf{E})$ that is sampled using three strings $s, \alpha(s)$ and $\beta(s)$ as a random source. Then $\mathrm{Fake}(\gamma, E) \cap \mathrm{Fake}(\gamma_\alpha, E_\alpha) \cap \mathrm{Fake}(\gamma_\beta, E_\beta) = \varnothing$.

Consider arbitrary strings $r \in R$ and $s \in S$. The first property implies that for random variables sampled using strings (r, s), $(r, \alpha(s))$ and $(r, \beta(s))$ as a random

source values of $P(\mathbf{P}_1, \ldots, \mathbf{P}_k)$ are the same. The second property implies that for at least one of this cases this value does not belong to $\mathrm{Fake}(\boldsymbol{\gamma}, \mathbf{E})$. Then, using that α and β are bijections, we get $\Pr[P(\mathbf{P}_1, \ldots, \mathbf{P}_k) \in \mathrm{Fake}(\boldsymbol{\gamma}, \mathbf{E})] \leq 2/3$. Since we have many random variables, it is a tedious task to construct such α and β. In order to simplify this task we slightly relax the properties. We will define bijections α and β not on all strings S but only on the part of bits corresponding to sampling of $\boldsymbol{\gamma}$ and $\mathbf{E}$. More precisely we will define two bijections α and β on the set of values of the random variable $(\boldsymbol{\gamma}, \mathbf{E})$. We relax the first property as follows:

1'. For every γ and E the three conditional distributions of the random variable $(\mathbf{P}_1, \ldots, \mathbf{P}_k)$ under the following three conditions coincide:

   (a) $(\boldsymbol{\gamma}, \mathbf{E}) = (\gamma, E)$,

   (b) $(\boldsymbol{\gamma}, \mathbf{E}) = \alpha(\gamma, E)$ and

   (c) $(\boldsymbol{\gamma}, \mathbf{E}) = \beta(\gamma, E)$.

Unfortunately, we were not able to construct such bijections on the set of all pairs (γ, E). Thus we take a set Ξ consisting $1 - \delta$ fraction of all values of $(\boldsymbol{\gamma}, \mathbf{E})$ and we will claim that α and β are bijections on Ξ. Such relaxation will weaken the bound of the probability up to $2/3 + \delta$. We formalize the requirements to Ξ, α and β in Definition24. Then we verify in Claim25 that these requirements are sufficient to bound $\Pr[P(\mathbf{P}_1, \ldots, \mathbf{P}_k) \in \mathrm{Fake}(\boldsymbol{\gamma}, \mathbf{E})]$. The construction of Ξ, α and β is given in Subsection5.3.

Definition 24. For a 0-instance $\vec{x}$ of $\mathrm{UDISJ}_{k, 2^{\ell-k}-1}$ and constant $1 > \delta \geq 0$, let Ξ consist of pairs (γ, E) with $\gamma : [2^{\ell-k} - 1] \cup \{*\} \to \{0,1\}^{\ell-k}$ bijective and $E \in \mathbb{F}_2^{\ell \times \ell}$ invertible. Bijections $\alpha, \beta : \Xi \to \Xi$ form a $(1-\delta)$-symmetry randomness space if:

- (Largeness) $\Pr[(\boldsymbol{\gamma}, \mathbf{E}) \in \Xi] \geq 1 - \delta$.
- (Difference) $\mathrm{Fake}(\gamma, E) \cap \mathrm{Fake}(\alpha(\gamma, E)) \cap \mathrm{Fake}(\beta(\gamma, E)) = \varnothing$ for every $(\gamma, E) \in \Xi$.
- (Symmetry) $C_{\vec{x}}(\gamma, E)$, $C_{\vec{x}}(\alpha(\gamma, E))$, $C_{\vec{x}}(\beta(\gamma, E))$ differ only by row permutation.

Claim 25. Under Definition24, $\Pr[P(\mathbf{P}_1, \ldots, \mathbf{P}_k) \in \mathrm{Fake}(\boldsymbol{\gamma}, \mathbf{E})] \leq 2/3 + \delta$.

Proof. Write $\vec{\boldsymbol{P}} = (\mathbf{P}_1, \ldots, \mathbf{P}_k)$, $\vec{\boldsymbol{\Delta}} = (\boldsymbol{\Delta}_1, \ldots, \boldsymbol{\Delta}_k)$, $\vec{D}(\vec{x}, \gamma) = (D_1(x_1, \gamma), D_2(x_2), \ldots, D_k(x_k))$, so $\vec{\boldsymbol{P}} = \vec{\boldsymbol{\Delta}} + M_{\boldsymbol{\pi}}(\vec{D}(\vec{x}, \boldsymbol{\gamma})\mathbf{E})$. Let $p = \Pr[P(\vec{\boldsymbol{P}}) \in \mathrm{Fake}(\boldsymbol{\gamma}, \mathbf{E})]$. Then

$$\begin{aligned} p &= \sum_{\gamma, E} \Pr[P(\vec{\boldsymbol{\Delta}} + M_{\boldsymbol{\pi}}(\vec{D}(\vec{x}, \gamma)E)) \in \mathrm{Fake}(\gamma, E)] \Pr[\boldsymbol{\gamma} = \gamma, \mathbf{E} = E] \\ &\leq \sum_{(\gamma, E) \in \Xi} \Pr[P(\vec{\boldsymbol{\Delta}} + M_{\boldsymbol{\pi}}(\vec{D}(\vec{x}, \gamma)E)) \in \mathrm{Fake}(\gamma, E)] \Pr[\boldsymbol{\gamma} = \gamma, \mathbf{E} = E] + \delta. \end{aligned}$$

Notice that for fixed γ, E the random variable $\vec{\boldsymbol{\Delta}} + M_{\boldsymbol{\pi}} \cdot (\vec{D}(\vec{x}, \gamma) \cdot E)$ is distributed uniformly on the set of tuples $(L_1, \ldots, L_k)$ of k matrices from $\mathbb{F}_2^{(2^\ell + 2^k) \times \ell}$

such that $\sum_{i=1}^{k} L_i$ differs from $C_{\vec{x}}(\gamma, E)$ only by a permutation of rows. Let $(\gamma_{\alpha^{-1}}, E_{\alpha^{-1}}) = \alpha^{-1}(\gamma, E)$. By the symmetry condition, matrices $C_{\vec{x}}(\gamma, E)$ and $C_{\vec{x}}(\gamma_{\alpha^{-1}}, E_{\alpha^{-1}})$ differ only by permutation of rows. Thus, for every set A the probability

$$\Pr[P(\vec{\boldsymbol{\Delta}} + M_{\boldsymbol{\pi}}(\vec{D}(\vec{x}, \gamma)E)) \in A] = \Pr[P(\vec{\boldsymbol{\Delta}} + M_{\boldsymbol{\pi}}(\vec{D}(\vec{x}, \gamma_{\alpha^{-1}})E_{\alpha^{-1}})) \in A].$$

Therefore

$$\begin{aligned}
p &\leq \sum_{(\gamma,E)\in\Xi} \Pr[P(\vec{\boldsymbol{\Delta}} + M_{\boldsymbol{\pi}}(\vec{D}(\vec{x}, \gamma_{\alpha^{-1}})E_{\alpha^{-1}})) \in \mathrm{Fake}(\gamma, E)] \Pr[\boldsymbol{\gamma} = \gamma, \mathbf{E} = E] + \delta \\
&= \sum_{(\gamma,E)\in\Xi} \Pr[P(\vec{\boldsymbol{\Delta}} + M_{\boldsymbol{\pi}}(\vec{D}(\vec{x}, \gamma)E)) \in \mathrm{Fake}(\alpha(\gamma, E))] \Pr[(\boldsymbol{\gamma}, \mathbf{E}) = \alpha(\gamma, E)] + \delta \\
&= \sum_{(\gamma,E)\in\Xi} \Pr[P(\vec{\boldsymbol{\Delta}} + M_{\boldsymbol{\pi}}(\vec{D}(\vec{x}, \gamma)E)) \in \mathrm{Fake}(\alpha(\gamma, E))] \Pr[(\boldsymbol{\gamma}, \mathbf{E}) = (\gamma, E)] + \delta \\
&= \Pr[P(\vec{\boldsymbol{P}}) \in \mathrm{Fake}(\alpha(\boldsymbol{\gamma}, \mathbf{E})), (\boldsymbol{\gamma}, \mathbf{E}) \in \Xi] + \delta.
\end{aligned}$$

Analogously, $p \leq \Pr[P(\vec{\boldsymbol{P}}) \in \mathrm{Fake}(\beta(\boldsymbol{\gamma}, \mathbf{E})), (\boldsymbol{\gamma}, \mathbf{E}) \in \Xi] + \delta$. Also the inequality $p \leq \Pr[P(\vec{\boldsymbol{P}}) \in \mathrm{Fake}(\boldsymbol{\gamma}, \mathbf{E}), (\boldsymbol{\gamma}, \mathbf{E}) \in \Xi] + \delta$ follows by the largeness condition. Then,

$$\begin{aligned}
3(1-p) &\geq \Pr[P(\vec{\boldsymbol{P}}) \notin \mathrm{Fake}(\beta(\boldsymbol{\gamma}, \mathbf{E})) \vee (\boldsymbol{\gamma}, \mathbf{E}) \notin \Xi] \\
&\quad + \Pr[P(\vec{\boldsymbol{P}}) \notin \mathrm{Fake}(\alpha(\boldsymbol{\gamma}, \mathbf{E})) \vee (\boldsymbol{\gamma}, \mathbf{E}) \notin \Xi] \\
&\quad + \Pr[P(\vec{\boldsymbol{P}}) \notin \mathrm{Fake}(\boldsymbol{\gamma}, \mathbf{E}) \vee (\boldsymbol{\gamma}, \mathbf{E}) \notin \Xi] - 3\delta \\
&\geq \Pr[P(\vec{\boldsymbol{P}}) \notin \mathrm{Fake}(\boldsymbol{\gamma}, \mathbf{E}) \cap \mathrm{Fake}(\alpha(\boldsymbol{\gamma}, \mathbf{E})) \cap \mathrm{Fake}(\beta(\boldsymbol{\gamma}, \mathbf{E})) \vee (\boldsymbol{\gamma}, \mathbf{E}) \notin \Xi] - 3\delta \\
&= 1 - 3\delta.
\end{aligned}$$

The last equality follows by the difference condition. Hence, $3(1-p) \geq 1 - 3\delta$, thus $p \leq 2/3 + \delta$. □

Lemma 26. For every 0-instance $\vec{x}$ of $\mathrm{UDISJ}_{k,2^{\ell-k}-1}$, for some $\delta < 1/3 - 1/100$ there is a $(1-\delta)$-symmetry randomness space.

Lemma26 (proved in Section5.3) and Claim25 imply for a constant $\varepsilon > 0$ that

$$\Pr[P(\mathbf{P}_1, \ldots, \mathbf{P}_k) \notin \mathrm{Fake}(\boldsymbol{\gamma}, \mathbf{E})] \geq \varepsilon + 1/100, \quad \Pr[P(\mathbf{P}_1, \ldots, \mathbf{P}_k) \notin \mathrm{Fake}(\boldsymbol{\gamma}, \mathbf{E}) \cup \{\perp\}] \geq \varepsilon.$$

Then, for $N = O(\log(1/\varepsilon))$, T gives a correct answer for every 0-instance with probability at least 2/3. □

5.3 Constructions of Ξ, α, β

Proof of Lemma 26. Assume that $x_1, \ldots, x_k$ is a 0-instance $\mathrm{UDISJ}_{k,2^{\ell-k}-1}$. Let $i_0 \in [2^{\ell-k}-1]$ be such that $x_{1,i_0} = x_{2,i_0} = \cdots = x_{k,i_0} = 1$. Hereinafter γ denotes a bijection from $[2^{\ell-k}-1] \cup \{*\}$ to $\{0,1\}^{\ell-k}$, E denotes an invertible matrix from $\mathbb{F}_2^{\ell\times\ell}$ and $e_1, e_2, \ldots, e_\ell$ denote rows of E. Before presenting constructions

of Ξ, α, and β we explain how we are going to establish symmetry and difference properties from Definition24. For every $s \in \{0,1\}^{\ell-k}$ and $b \in \{0,1\}$ we introduce the following notation:

$$R(s,b,E) := \{(s,b,z)E \mid z \in \mathbb{F}_2^{k-1}\}.$$

Claim 27. The last 2^{k+1} rows of $C_{\vec{x}}(\gamma, E)$ form $R(\gamma(*),0,E) \cup R(\gamma(*),1,E)$, each occurring twice. This is Fake(γ, E). The rows with indices $[2^k(i-1)+1; 2^k i]$ for $i \in [2^{\ell-k-1}] \setminus \{i_0\}$ are $R(\gamma(i),0,E) \cup R(\gamma(i),1,E)$, each occurring once. The rows with indices $[2^k(i_0-1)+1; 2^k i_0]$ are $R(\gamma(i_0),0,E)$, each occurring twice.

Claim 28. $R(s,b,E) = (\sum_{j=1}^{\ell-k} s_j e_j + b e_{\ell-k+1}) + \mathrm{Span}(e_{\ell-k+2}, \ldots, e_\ell)$.

Proof.

$$\begin{aligned} R(s,b,E) &= \{(s,b,z)(e_1,\ldots,e_\ell)^T \mid z \in \mathbb{F}_2^{k-1}\} \\ &= \left\{ \sum_{j=1}^{\ell-k} s_j e_j + b e_{\ell-k+1} + \sum_{i=1}^{k-1} z_i e_{\ell-k+1+i} \mid z \in \mathbb{F}_2^{k-1} \right\} \\ &= \left(\sum_{j=1}^{\ell-k} s_j e_j + b e_{\ell-k+1} \right) + \mathrm{Span}(e_{\ell-k+2}, \ldots, e_\ell). \end{aligned}$$

□

Claim 29. $|R(s,b,E)| = 2^{k-1}$.

Proof. By Claim28, $|R(s,b,E)| = |(\sum_{j=1}^{\ell-k} s_j e_j + b e_{\ell-k+1}) + \mathrm{Span}(e_{\ell-k+2}, \ldots, e_\ell)| = |\mathrm{Span}(e_{\ell-k+2}, \ldots, e_\ell)| = 2^{k-1}$. □

Claim 30. The sets $R(s,b,E)$ for $s \in \{0,1\}^{\ell-k}, b \in \{0,1\}$ are disjoint.

Proof. Consider two vectors $u \in R(s,b,E)$ and $v \in R(s',b',E)$ such that $(s,b) \neq (s',b')$. Then, by Claim28, u and v have different coordinates in the basis $e_1, e_2, \ldots, e_\ell$, hence $u \neq v$. □

Claim 31. For bijections γ, γ' and invertible E, E', if

$$\begin{aligned} &R(\gamma(i_0),0,E) \cup R(\gamma(*),0,E) \cup R(\gamma(*),1,E) \\ &\quad = R(\gamma'(i_0),0,E') \cup R(\gamma'(*),0,E') \cup R(\gamma'(*),1,E'), \\ &R(\gamma(i_0),1,E) = R(\gamma'(i_0),1,E'), \end{aligned}$$

then $C_{\vec{x}}(\gamma, E)$ and $C_{\vec{x}}(\gamma', E')$ differ only by a row permutation.

Proof. By Claim27, rows from $R(\gamma(i_0), 1, E)$ do not appear in $C_{\vec{x}}(\gamma, E)$, rows from $R(\gamma(i_0), 0, E) \cup R(\gamma(*), 0, E) \cup R(\gamma(*), 1, E)$ appear in $C_{\vec{x}}(\gamma, E)$ exactly twice. The matrix $C_{\vec{x}}(\gamma, E)$ has $2^\ell + 2^k$ rows. All rows of $C_{\vec{x}}(\gamma, E)$ that are not in $R(\gamma(i_0), 1, E) \cup R(\gamma(*), 0, E) \cup R(\gamma(*), 1, E)$, by Claim27, appear in $C_{\vec{x}}(\gamma, E)$ exactly once. By Claims29 and30, $|R(\gamma(i_0), 0, E) \cup R(\gamma(*), 0, E) \cup R(\gamma(*), 1, E)| = 3 \cdot 2^{k-1}$, hence, the number of rows of $C_{\vec{x}}(\gamma, E)$ that are not in $R(\gamma(i_0), 1, E) \cup R(\gamma(*), 0, E) \cup R(\gamma(*), 1, E)$ equals $2^\ell - 2^{k+1}$. By Claims29 and30, the number of ℓ-bit strings not from $R(\gamma(i_0), 1, E) \cup R(\gamma(i_0), 0, E) \cup R(\gamma(*), 0, E) \cup R(\gamma(*), 1, E)$ is also $2^\ell - 2^{k+1}$. Hence, all rows from $\{0,1\}^\ell \setminus (R(\gamma(i_0), 0, E) \cup R(\gamma(*), 0, E) \cup (\gamma(*), 1, E) \cup R(\gamma(i_0), 1, E))$ appear in $C_{\vec{x}}(\gamma, E)$ exactly once. Thus, matrices $C_{\vec{x}}(\gamma, E)$ and $C_{\vec{x}}(\gamma', E')$ have the same set of rows and each row appears the same number of times in each of these matrices. □

Write $\alpha(\gamma, E) = (\gamma_\alpha, E_\alpha)$ and $\beta(\gamma, E) = (\gamma_\beta, E_\beta)$. We construct them so that

$$\begin{aligned} R(\gamma(i_0), 1, E) &= R(\gamma_\alpha(i_0), 1, E_\alpha) = R(\gamma_\beta(i_0), 1, E_\beta), \\ R(\gamma(i_0), 0, E) &= R(\gamma_\alpha(*), 0, E_\alpha) = R(\gamma_\beta(*), 0, E_\beta), \\ R(\gamma(*), 1, E) &= R(\gamma_\alpha(*), 1, E_\alpha) = R(\gamma_\beta(i_0), 0, E_\beta), \\ R(\gamma(*), 0, E) &= R(\gamma_\alpha(i_0), 0, E_\alpha) = R(\gamma_\beta(*), 1, E_\beta). \end{aligned} \tag{2}$$

Notice that by Claim31, equations(2) imply the symmetry property. Equations(2) also imply the difference property. Indeed, the three Fake sets are, respectively,

$$\begin{aligned} &R(\gamma(*), 1, E) \cup R(\gamma(*), 0, E), \\ &R(\gamma(*), 1, E) \cup R(\gamma(i_0), 0, E), \\ &R(\gamma(*), 0, E) \cup R(\gamma(i_0), 0, E), \end{aligned}$$

Hence, by Claim30, $\mathrm{Fake}(\gamma, E) \cap \mathrm{Fake}(\gamma_\alpha, E_\alpha) \cap \mathrm{Fake}(\gamma_\beta, E_\beta) = \varnothing$. It remains to construct Ξ, α, β satisfying (2) and $\Pr[(\boldsymbol{\gamma}, \mathbf{E}) \in \Xi] > 2/3 + 1/100$.

Definition of Ξ $(\gamma, E) \in \Xi$ iff there exist $m, n \in [\ell - k]$ with $\gamma(*)_m = 1, \gamma(i_0)_m = 0$ and $\gamma(*)_n = 0, \gamma(i_0)_n = 1$, i.e. $\gamma(*)$ and $\gamma(i_0)$ are coordinatewise incomparable. $\boldsymbol{\gamma}(i_0)$ and $\boldsymbol{\gamma}(*)$ are uniform distinct elements. For independent uniform random subsets $\mathbf{S}, \mathbf{T} \subseteq [\ell - k]$,

$$\begin{aligned} \Pr[(\boldsymbol{\gamma}, \mathbf{E}) \in \Xi] &= 1 - \Pr[\boldsymbol{\gamma}(i_0) \le \boldsymbol{\gamma}(*) \vee \boldsymbol{\gamma}(*) \le \boldsymbol{\gamma}(i_0)] \\ &\ge 1 - 2\Pr[\boldsymbol{\gamma}(i_0) \le \boldsymbol{\gamma}(*)] \\ &= 1 - 2\Pr[\mathbf{S} \subseteq \mathbf{T} \mid \mathbf{S} \ne \mathbf{T}] \\ &\ge 1 - 2\Pr[\mathbf{S} \subseteq \mathbf{T}] \\ &= 1 - 2\prod_{j=1}^{\ell-k}(1 - \Pr[j \in \mathbf{S} \wedge j \notin \mathbf{T}]) \\ &= 1 - 2(3/4)^{\ell-k} > 2/3 + 1/100 \qquad (\ell - k \ge 7). \end{aligned}$$

Hence largeness holds.

Construction of α Let E_α have rows

$$(e'_1, \dots, e'_\ell) = (e_1, \dots, e_{\ell-k}, e_{\ell-k+1} + \sum_{j=1}^{\ell-k} (\gamma(i_0)_j + \gamma(*)_j) e_j, e_{\ell-k+2}, \dots, e_\ell),$$

and $\gamma_\alpha(i) = \gamma(*)$ for $i = i_0$, $\gamma(i_0)$ for $i = *$, and $\gamma(i)$ otherwise.

Claim 32. $\alpha : \Xi \to \Xi$ is bijective.

Proof. Notice that rows of E' form a basis since $\sum_{j=1}^{\ell-k} (\gamma(i_0)_j + \gamma(*)_j) e_j \in \mathrm{Span}(e_1, \dots, e_{\ell-k})$. The mapping $\gamma \mapsto \gamma_\alpha$ is bijective since it just swaps $\gamma(i_0)$ and $\gamma(*)$. Since the condition on $\gamma(i_0)$ and $\gamma(*)$ does not change after application of α, we get that $\alpha(\Xi) \subseteq \Xi$. Notice that

$$\sum_{j=1}^{\ell-k} (\gamma(i_0)_j + \gamma(*)_j) e_j = \sum_{j=1}^{\ell-k} (\gamma_\alpha(i_0)_j + \gamma_\alpha(*)_j) e'_j,$$

hence $\alpha(\gamma_\alpha, E_\alpha) = (\gamma, E)$, hence α is bijective. □

Claim 33. For all $(\gamma, E) \in \Xi$,

$$\begin{aligned} R(\gamma_\alpha(i_0), 1, E_\alpha) &= R(\gamma(i_0), 1, E), \\ R(\gamma_\alpha(i_0), 0, E_\alpha) &= R(\gamma(*), 0, E), \\ R(\gamma_\alpha(*), 0, E_\alpha) &= R(\gamma(i_0), 0, E), \\ R(\gamma_\alpha(*), 1, E_\alpha) &= R(\gamma(*), 1, E). \end{aligned}$$

Proof. Use Claim28 and $S := \mathrm{Span}(e_{\ell-k+2}, \dots, e_\ell) = \mathrm{Span}(e'_{\ell-k+2}, \dots, e'_\ell)$:

$$\begin{aligned} R(\gamma_\alpha(i_0), 1, E_\alpha) &= (\sum_j \gamma_\alpha(i_0)_j e'_j + e'_{\ell-k+1}) + S \\ &= (\sum_j \gamma(*)_j e'_j + e'_{\ell-k+1}) + S = (\sum_j \gamma(i_0)_j e_j + e_{\ell-k+1}) + S = R(\gamma(i_0), 1, E), \\ R(\gamma_\alpha(i_0), 0, E_\alpha) &= (\sum_j \gamma_\alpha(i_0)_j e'_j) + S = (\sum_j \gamma(*)_j e_j) + S = R(\gamma(*), 0, E), \\ R(\gamma_\alpha(*), 0, E_\alpha) &= (\sum_j \gamma_\alpha(*)_j e'_j) + S = (\sum_j \gamma(i_0)_j e_j) + S = R(\gamma(i_0), 0, E), \\ R(\gamma_\alpha(*), 1, E_\alpha) &= (\sum_j \gamma_\alpha(*)_j e'_j + e'_{\ell-k+1}) + S \\ &= (\sum_j \gamma(i_0)_j e'_j + e'_{\ell-k+1}) + S = (\sum_j \gamma(*)_j e_j + e_{\ell-k+1}) + S = R(\gamma(*), 1, E). \end{aligned}$$

Here each sum is over $j = 1, \dots, \ell - k$. □

Construction of β Set $\gamma_\beta = \gamma_\alpha$. Define $j_{\min} = \min\{j \in [\ell - k] : \gamma(*)_j = 1 \wedge \gamma(i_0)_j = 0\}$, which exists on Ξ. Set $E_\beta = (e''_1, \dots, e''_\ell)$ with

$$e''_j = \begin{cases} e_j, & j \notin \{j_{\min}, \ell - k + 1\}, \\ \sum_{i=1}^{\ell-k} (\gamma(*)_i + \gamma(i_0)_i) e_i, & j = \ell - k + 1, \\ e_{j_{\min}} + e_{\ell-k+1}, & j = j_{\min}. \end{cases}$$

Claim 34. $\beta : \Xi \to \Xi$ is bijective.

Proof. Let us verify that β is injective. Given γ_β we can easily recover γ, hence we can recover $j_{\min}$ as well. Then

$$\begin{aligned} \sum_{i=1}^{\ell-k} (\gamma(i_0)_i + \gamma(*)_i) e''_i + e''_{\ell-k+1} &= \sum_{i \in [\ell-k] \setminus \{j_{\min}\}} (\gamma(i_0)_i + \gamma(*)_i) e_i + e_{j_{\min}} + e_{\ell-k+1} + e''_{\ell-k+1} \\ &= e_{\ell-k+1} + \underbrace{\sum_{i \in [\ell-k] \setminus \{j_{\min}\}} (\gamma(i_0)_i + \gamma(*)_i) e_i + e_{j_{\min}}}_{e''_{\ell-k+1}} + e''_{\ell-k+1} \\ &= e_{\ell-k+1}. \end{aligned}$$

Thus, we can uniquely recover $e_{\ell-k+1}$ and, hence, also recover $e_{j_{\min}} = e''_{j_{\min}} + e_{\ell-k+1}$; for $j \in [\ell] \setminus \{j_{\min}, \ell - k + 1\}$, $e_j = e''_j$. Hence, β is injective. Notice that since we represent $e_1, \dots, e_\ell$ as linear combinations of $e''_1, \dots, e''_\ell$, then $e''_1, \dots, e''_\ell$ is a basis, hence the matrix E_β is invertible. Thus, we verify that $\beta(\Xi) \subseteq \Xi$ and β is injective, hence β is bijective. □

Claim 35. For all $(\gamma, E) \in \Xi$,

$$\begin{aligned} R(\gamma_\beta(i_0), 1, E_\beta) &= R(\gamma(i_0), 1, E), \\ R(\gamma_\beta(i_0), 0, E_\beta) &= R(\gamma(*), 1, E), \\ R(\gamma_\beta(*), 0, E_\beta) &= R(\gamma(i_0), 0, E), \\ R(\gamma_\beta(*), 1, E_\beta) &= R(\gamma(*), 0, E). \end{aligned}$$

Proof. Put $S = \mathrm{Span}(e_{\ell-k+2}, \dots, e_\ell) = \mathrm{Span}(e''_{\ell-k+2}, \dots, e''_\ell)$. Recall $\gamma(*)_{j_{\min}} = 1$, $\gamma(i_0)_{j_{\min}} = 0$. Sums below run from1 to $\ell - k$:

$$\begin{aligned} R(\gamma_\beta(i_0), 1, E_\beta) &= \sum_i \gamma_\beta(i_0)_i e''_i + e''_{\ell-k+1} + S \\ &= \sum_i \gamma(*)_i e_i + e_{\ell-k+1} + e''_{\ell-k+1} + S \\ &= \sum_i \gamma(*)_i e_i + e_{\ell-k+1} + \sum_i (\gamma(*)_i + \gamma(i_0)_i) e_i + S \\ &= \sum_i \gamma(i_0)_i e_i + e_{\ell-k+1} + S = R(\gamma(i_0), 1, E), \\ R(\gamma_\beta(i_0), 0, E_\beta) &= \sum_i \gamma_\beta(i_0)_i e''_i + S = \sum_i \gamma(*)_i e_i + e_{\ell-k+1} + S = R(\gamma(*), 1, E), \end{aligned}$$

$$R(\gamma_\beta(*),0,E_\beta)=\sum_i \gamma_\beta(*)_i e''_i + S = \sum_i \gamma(i_0)_i e_i + S = R(\gamma(i_0),0,E),$$

$$R(\gamma_\beta(*),1,E_\beta)=\sum_i \gamma_\beta(*)_i e''_i + e''_{\ell-k+1} + S = \sum_i \gamma(i_0)_i e_i + e''_{\ell-k+1} + S = \sum_i \gamma(*)_i e_i + S = R(\gamma(*),0,E).$$

□

Claims33 and35 imply equations(2). □

5.4 Proof of Lemma 21

To prove Lemma21 it is sufficient to prove:

Proposition 36. There exist matrices $T_1,\dots,T_k \in \mathbb{F}_2^{2^k\times k}$ such that (1) for $\alpha_i \in \{0,1\}$, $\sum_i \alpha_i T_i = 0$ iff all $\alpha_i = 0$, i.e. they are linearly independent; (2) every nonzero $M \in \mathrm{Span}(T_1,\dots,T_k)$ satisfies $M + K_k \in S_k$.

Proof of Lemma21. Let for $i \in \{1,\dots,k-1\}$, $A_i(0) = T_i$ and $A_i(1)$ be the zero matrix. Let $A_k(0) = K_k + T_k$, $A_k(1) = K_k$. For each $b_1,\dots,b_k \in \{0,1\}$, $\sum_{i=1}^k A_i(b_i) = \sum_{i=1}^k (1-b_i)T_i + K_k$. Then $\sum_{i=1}^k A_i(1) = K_k$, and if for at least one $i \in [k]$, $b_i \neq 1$, then by the first condition of Proposition36, $\sum_{i=1}^k (1-b_i)T_i$ differs from zero, thus by the second condition of Proposition36, $\sum_{i=1}^k A_i(b_i) \in S_k$. □

Proof of Proposition36. Let us prove the proposition by induction on k. We are going to prove a stronger statement: namely, we additionally require that for arbitrary non-zero matrix $M \in \mathrm{Span}(T_1,\dots,T_k)$ the set of even-indexed rows of $M + K_k \in S_k$ coincide with the set of odd-indexed rows of this matrix with all bits flipped. The base case $k = 1$. $T_1 = \binom{0}{1}$, and $K_1 = \binom{0}{0}$. It is easy to verify that all conditions hold. Induction step from k to $k+1$. Notice that

$$K_{k+1} = \begin{pmatrix} K_k & 0_{2^k\times 1} \\ K_k & 1_{2^k\times 1} \end{pmatrix}.$$

Let $T_1,\dots,T_k$ be the matrices from induction hypothesis for k. Then define

$$T'_i = \begin{pmatrix} T_i & 0_{2^k\times 1} \\ T_i & 0_{2^k\times 1} \end{pmatrix}$$

for $i \in [k]$ and

$$T'_{k+1} = \begin{pmatrix} 0_{2^k\times k} & z_0 \\ 1_{2^k\times k} & z_1 \end{pmatrix},$$

where $z_0 = (0,1,0,1,\dots,0,1)^T \in \{0,1\}^{2^k\times 1}$, and $z_1 = (1,0,1,0,\dots,1,0)^T \in \{0,1\}^{2^k\times 1}$. Let us verify that all conditions hold. First we show that the matrices $T'_1, T'_2, \dots, T'_{k+1}$ are linearly independent. Matrices $T'_1, T'_2, \dots, T'_k$ are linearly independent since they contain linearly independent blocks $T_1,\dots,T_k$. The matrix T'_{k+1} does not belong to $\mathrm{Span}(T'_1,\dots,T'_k)$, since the last column of T'_{k+1} is non-zero, but the last columns of all $T'_1,\dots,T'_k$ are zeros. Let us check that for any non-zero matrix $M \in \mathrm{Span}(T'_1,\dots,T'_k,T'_{k+1})$, the condition $M + K_{k+1} \in S_{k+1}$ holds and the set of even-indexed rows of $M + K_{k+1}$ coincide with the set of odd-indexed rows of this matrix with all bits flipped. Let us analyze the cases:

1. Let M be a non-zero matrix from $\mathrm{Span}(T'_1, \ldots, T'_k)$. Then, M has form

$$\begin{pmatrix} M' & 0_{2^k \times 1} \\ M' & 0_{2^k \times 1} \end{pmatrix},$$

where M' is a non-zero matrix from $\mathrm{Span}(T_1, \ldots, T_k)$, thus $M' + K_k \in S_k$. Then

$$M + K_{k+1} = \begin{pmatrix} M' + K_k & 0_{2^k \times 1} \\ M' + K_k & 1_{2^k \times 1} \end{pmatrix};$$

it follows from the induction hypothesis that all rows of this matrix are distinct, i.e. $M + K_{k+1} \in S_{k+1}$. In order to verify that the set of even-indexed rows of this matrix coincide with the set of odd-indexed rows with all bits flipped, observe that by induction hypothesis the first 2^{k-1} even-indexed rows of $M + K_{k+1}$ coincide with the last 2^{k-1} odd-indexed rows of $M + K_{k+1}$ with all bits flipped, and the first 2^{k-1} odd-indexed rows of $M + K_{k+1}$ coincide with the last 2^{k-1} even-indexed rows of $M + K_{k+1}$ with flipped bits.

2. $M = T'_{k+1}$, then

$$M + K_{k+1} = \begin{pmatrix} K_k & z_0 \\ 1_{2^k \times k} + K_k & z_0 \end{pmatrix}.$$

Let us show that all rows of this matrix are distinct. The first 2^k rows start with0 and are obtained by appending zeroes and ones to the rows of K_k in the alternating order. Since for every pair of coinciding rows of K_k they are adjacent, the first 2^k rows are distinct. The last 2^k rows start from one, so they differ from the first 2^k rows. The proof that they are distinct is the same as for the first 2^k rows. Observe that the $(2i-1)$th row of the matrix $M + K_{k+1}$ coincide with the $(2^k + 2i)$th row of $M + K_{k+1}$ with flipped bits, and the $(2i)$th row of $M + K_{k+1}$ coincide with the $(2^k + 2i - 1)$th row of $M + K_{k+1}$ with flipped bits for $i \in [2^k]$.

3. $M = R + T'_{k+1}$, where R is a non-zero matrix from $\mathrm{Span}(T'_1, \ldots, T'_k)$. Let R have the form

$$\begin{pmatrix} R' & 0_{2^k \times 1} \\ R' & 0_{2^k \times 1} \end{pmatrix},$$

where R' is a non-zero matrix from $\mathrm{Span}(T_1, \ldots, T_k)$. Then

$$M + K_{k+1} = R + T'_{k+1} + K_{k+1} = \begin{pmatrix} R' + K_k & z_0 \\ 1_{2^k \times k} + R' + K_k & z_0 \end{pmatrix}.$$

By the induction hypothesis, $R' + K_k \in S_k$ and its even-indexed rows coincide with its odd-indexed rows with flipped bits. Then, all even-indexed rows of $M + K_{k+1}$ end with0, the first 2^{k-1} of them are even-indexed rows of $R' + K_k$ with appended zero, and the last 2^{k-1} of them are even-indexed rows of $R' + K_k$ with all bits flipped and appended0. Then, by the induction hypothesis, the set of the former rows does not intersect with the set of the latter rows, therefore they are all distinct. By the same argument, all the rows of $M + K_{k+1}$ that end with1 are distinct. Thus, $M + K_{k+1} \in S_{k+1}$. Let us verify that the set of even-indexed rows

of this matrix coincide with the set of odd-indexed rows of this matrix with all bits flipped. Observe that if the ith row of $R' + K_k$ coincides with the jth row of $R' + K_k$ with flipped bits, then the ith row of $M + K_{k+1}$ coincides with its jth row with flipped bits, and the $(2^k + i)$th row of $M + K_{k+1}$ coincides with its $(2^k + j)$th row with all bits flipped. The required property follows from the induction hypothesis.

□

5.5 Corollaries

Corollary37. If $k + 7 \le \ell$, the size of any semantic $\mathrm{Res}(\mathrm{PC}_{k-1})$ tree-like refutation of $\mathrm{BPHP}^{2^\ell+2^k}_{2^\ell}$ is at least $2^{\Omega(2^{\ell/2}/(k2^{3k/2}))}$. For $k = 2$, every tree-like semantic $\mathrm{Res}(\oplus)$ refutation of $\mathrm{BPHP}^{2^\ell+4}_{2^\ell}$ has size at least $2^{\Omega(2^\ell)}$.

Proof. Follows from Theorem3 and Lemma1. □

Corollary38. Let $2 \le k \le \ell - 7$ and S be the minimal size of tree-like refutation of $\phi = \mathrm{BPHP}^{2^\ell+2^k}_{2^\ell} \circ \oplus_k$ in the semantic proof system $\mathrm{T}^{\mathrm{cc}}(k, c)$. Then $\log S \log\log S \ge c \cdot \Omega(2^{\ell/2}/(k2^{3k/2}))$. For $k = 2$, $\log S \log\log S \ge c \cdot \Omega(2^\ell)$.

Proof. By Lemma9, $R^{1/3}_{\mathrm{pub}}(\mathrm{Search}(\phi)) = O(\log S \log\log S/c)$. Also

$$R^{1/3}_{\mathrm{pub}}(\mathrm{Search}(\mathrm{BPHP}^{2^\ell+2^k}_{2^\ell} \circ \oplus_k)) \ge R^{1/3}_{\mathrm{pub}}(\oplus_k \mathrm{Search}(\mathrm{BPHP}^{2^\ell+2^k}_{2^\ell})).$$

Now apply Theorem3. □

6 Bit pigeonhole principle

6.1 Reduction from $\mathrm{BPHP} \circ \oplus_k$ to BPHP

Let $T \subseteq X_1 \times \cdots \times X_k \times Y$ and $S \subseteq Z_1 \times \cdots \times Z_k \times W$ be relations. We say S is many-one reducible to T if there are maps $f_i : X_i \to Z_i$ and $g : W \to Y$ such that if $(f_1(x_1), \ldots, f_k(x_k), y) \in T$ then $(x_1, \ldots, x_k, g(y)) \in S$.

Lemma39. If S is many-one reducible to T, then $R^{\mathrm{pub}}_{1/3}(S) \le R^{\mathrm{pub}}_{1/3}(T)$.

Proof. The ith party computes $f(x_j)$ for all $j \in [k] \setminus \{i\}$ and then all parties run the optimal protocol for T. As soon as all the parties learn an answer y they compute $g(y)$ without communication. □

Recall $\mathrm{BPHP}^M_{2^n}$ encodes M different strings $s_1, \ldots, s_M \in \{0,1\}^n$. For $n = \ell k + r$, $0 \le r < k$, partition each row $s_i = s_i^{(1)} \cdots s_i^{(k)}$, where $|s_i^{(t)}| = \ell + 1$ for $t \le r$ and ℓ for $t > r$. Partition Π_k puts all $s_1^{(t)}, \ldots, s_M^{(t)}$ in part t. $\mathrm{SearchPair}^M_{2^n}$ asks for distinct i, j with $s_i = s_j$ from variables partitioned according to Π_k.

Proposition40. $\mathrm{SearchPair}^M_{2^n}$ is many-one reducible to $\mathrm{Search}(\mathrm{BPHP}^M_{2^n})$ with partition Π_k.

Proof. The proof is straightforward. □

Theorem41. $\oplus_k\mathrm{BPHP}^m_{2^\ell}$ is many-one reducible to $\mathrm{SearchPair}^{m2^{(k-1)\ell}}_{2^{k\ell}}$.

Proof. Let us denote $M = m \cdot 2^{(k-1)\ell}$. Consider a set $Z = \{(y_1, y_2, \ldots, y_k) \in (\mathbb{F}_2^\ell)^k \mid \sum_i y_i = 0\}$. It is easy to see that $|Z| = 2^{(k-1)\ell}$. Let ϕ be a bijection between $[M]$ and $Z \times [m]$. Let for $i \in [m]$ and $t \in [k]$, $x_i^{(t)}$ denote the ith string of the tth party in the communication problem $\oplus_k\mathrm{BPHP}^m_{2^\ell}$. Let $x_i := (x_i^{(1)}, \ldots, x_i^{(k)})$. For every $t \in [k]$ we define f_t as follows: $f_t(x_1^{(t)}, \ldots, x_m^{(t)})$ is a sequence of rows $r_1^{(t)}, r_2^{(t)}, \ldots, r_M^{(t)}$ such that for all $i \in [M]$, $r_i^{(t)} = z_t + x_j^{(t)}$, where $(z, j) = \phi(i)$ for all $z \in Z$ and $j \in [m]$ (recall that $z \in Z$ is divided on k parts of equal lengths and z_t denotes the tth part). Let us construct the function g from the definition of the reduction. Let $q, w \in [M]$ and $q \neq w$. Assume that $\phi(q) = (z, j_1)$ and $\phi(w) = (z, j_2)$. We define $g(q, w) := (j_1, j_2)$. Let us verify that $f_1, f_2, \ldots, f_k$ and g define a reduction. Let $q, w \in M$ be a pair of different numbers such that the assignment $\alpha := \{s_i \leftarrow r_i^{(1)} r_i^{(2)} \cdots r_i^{(k)} \mid i \in [M]\}$ satisfies $s_q = s_w$. Assume that $g(q, w) = (j_1, j_2)$. We need to verify that $j_1 \neq j_2$ and $\sum_{t=1}^k x_{j_1}^{(t)} = \sum_{t=1}^k x_{j_2}^{(t)}$. Notice that under the assignment α the value of s_q is $x_{j_1} + z$ and the value of s_w is $x_{j_2} + y$, where $j_1, j_2 \in [m]$ and $z, y \in Z$ such that $(z, j_1) = \phi(q)$ and $(y, j_2) = \phi(w)$. If $j_1 = j_2$, then $x_{j_1} + z = x_{j_2} + y$ implies $z = y$. Since ϕ is a bijection, we get $q = w$. Thus, $j_1 \neq j_2$. For each $t \in [k]$, the following equality holds.

$$z_t + x_{j_1}^{(t)} = y_t + x_{j_2}^{(t)}. \tag{3}$$

If we sum up equations(3) for all $t \in [k]$ and use that $y, z \in Z$, we get $\sum_{t=1}^k x_{j_1}^{(t)} = \sum_{t=1}^k x_{j_2}^{(t)}$. Hence, (j_1, j_2) is a correct answer for $\oplus_k\mathrm{BPHP}^m_{2^\ell}$. □

The following proposition deals with the case, where the number of bits is not divisible by k.

Proposition42. For $n = k\ell + r$, $0 \leq r < k$, $M > 2^{k\ell}$, $\mathrm{SearchPair}^M_{2^{k\ell}}$ is many-one reducible to $\mathrm{SearchPair}^{M2^r}_{2^n}$.

Proof. Let $x_1, x_2, \ldots, x_M$ be the input of $\mathrm{SearchPair}^M_{2^{k\ell}}$, let $x_j^{(t)}$ be the tth part of the row x_j according to the partition Π_k. Given this input we construct an input for $\mathrm{SearchPair}^{M2^r}_{2^n}$. Let τ be a bijection between $[M] \times \{0,1\}^r$ and $[M2^r]$. For each $i \in [M]$ we construct 2^r rows $y_{\tau(i,\alpha)}$ one for each $\alpha \in \{0,1\}^r$. Let Π_k partition a row $y_{\tau(i,\alpha)}$ into the following parts: $y^{(1)}_{\tau(i,\alpha)} y^{(2)}_{\tau(i,\alpha)} \cdots y^{(k)}_{\tau(i,\alpha)}$. Let

$$y^{(t)}_{\tau(i,\alpha)} = \begin{cases} x_i^{(t)}, & t > r, \\ x_i^{(t)} \alpha_t, & 0 \leq t \leq r. \end{cases}$$

Now we can define the function $f_t(x_1^{(t)}, \ldots, x_M^{(t)})$ as $y^{(t)}_{\tau(i,\alpha)}$ for each $i \in M$ and $\alpha \in \{0,1\}^r$ and $t \in [k]$. Observe that for each $i \in [M]$ the rows $y_{i,\alpha}$ for $\alpha \in \{0,1\}^r$ are distinct. That allows us to define the function g as $g(\tau(i_1, \alpha_1), \tau(i_2, \alpha_2)) = (i_1, i_2)$. All the required properties can be easily verified. □

Theorem4. Let $M = 2^n + 2^{k+n-\lfloor n/k \rfloor}$ and $n \geq k(k+7)$. With partition Π_k,

$$R^{\mathrm{pub}}_{1/3}(\mathrm{Search}(\mathrm{BPHP}^M_{2^n})) = \Omega(2^{n/(2k)-3k/2}/k).$$

For $k = 2$ the stronger bound is $\Omega(2^{n/2})$.

Proof. Let $\ell = \lfloor n/k \rfloor$, $r = n - \ell k$. By Proposition40, Proposition42, Theorem41 and Corollary19, respectively,

$$\begin{aligned}
R^{\mathrm{pub}}_{1/3}(\mathrm{Search}(\mathrm{BPHP}^M_{2^n})) &= R^{\mathrm{pub}}_{1/3}(\mathrm{Search}(\mathrm{BPHP}^{(2^k+2^\ell)2^{(k-1)\ell+r}}_{2^n})) \\
&\geq R^{\mathrm{pub}}_{1/3}(\mathrm{SearchPair}^{(2^k+2^\ell)2^{(k-1)\ell+r}}_{2^n}) \\
&\geq R^{\mathrm{pub}}_{1/3}(\mathrm{SearchPair}^{(2^k+2^\ell)2^{(k-1)\ell}}_{2^{k\ell}}) \\
&\geq R^{\mathrm{pub}}_{1/3}(\oplus_k \mathrm{BPHP}^{2^\ell+2^k}_{2^\ell}) \\
&= \Omega(2^{\ell/2-3k/2}/k) = \Omega(2^{n/(2k)-3k/2}/k).
\end{aligned}$$

The case of $k = 2$ can be treated in the same way, the only difference is in the application of Corollary19. □

6.2 Upper bound for communication complexity

Proposition5. For $M > 2^n$ and $k \in \{2, \ldots, n\}$ there is a deterministic NOF protocol for $\mathrm{Search}(\mathrm{BPHP}^M_{2^n})$ with partition Π_k transmitting $O(2^{\lceil n/k \rceil} \log M)$ bits.

Proof. The protocol is going to have only two active parties: the second party, which we call Alice, and the first party, which we call Bob. We are going to use that Alice can see the variables $s^{(1)}_1, \ldots, s^{(1)}_M$ and that Bob can see all other variables. Let us denote $\bar{s}_i = s^{(1)}_i s^{(2)}_i s^{(3)}_i \cdots s^{(k)}_i \in \{0,1\}^{n-\lceil n/k \rceil}$ the bits Bob sees in the ith line for $i \in [M]$. Bob finds a value $\alpha \in \{0,1\}^{n-\lceil n/k \rceil}$ such that the size of the set $S_\alpha = \{i \in [M] \mid \bar{s}^{(1)}_i = \alpha\}$ is larger than $2^{\lceil n/k \rceil}$. Such α exists since $M > 2^n$. Bob then picks an arbitrary subset S' of S_α of size $2^{\lceil n/k \rceil} + 1$ and sends the description of S' to Alice using $(2^{\lceil n/k \rceil} + 1) \cdot \lceil \log_2 M \rceil$ bits. Then, by the pigeonhole principle there exists $i \neq j \in S'$ such that $s^{(1)}_i = s^{(1)}_j$. Alice and Bob then spend $O(\log M + n)$ bits transmitting indices i and j and all the values of the ith and jth lines to each other. Both of them then find the falsified clause of $\mathrm{BPHP}^M_{2^n}$ with no communication because it only depends on variables s_i and s_j and broadcast its description to all of the parties using an additional $O(n + \log M)$ bits. □

For $k = 2$ this coincides with the lower bound up to a logarithmic factor. For larger k they are polynomially related; the dependence on k is a genuine phenomenon.

6.3 Short $\mathrm{Th}(\log n)$ proof of BPHP_n^m

This gives a short tree-like $\mathrm{Th}(\log n)$ refutation, similar to the conversion of unary PHP resolution proofs to $\mathrm{Res}(\log n)$ proofs of binary PHP in [5].

Proposition43. If there exists a tree-like $\mathrm{Th}(1)$ refutation of $\mathrm{PHP}_{2^\ell}^m$ of size S, there exists a tree-like $\mathrm{Th}(\ell)$ refutation of $\mathrm{BPHP}_{2^\ell}^m$ of size $O(S)$.

Proof. Let $p_{i,j}$ for $i \in [m]$ and $j \in [2^\ell]$ be a variable of $\mathrm{PHP}_{2^\ell}^m$ indicating that the ith pigeon flies to the jth hole. Let $s_{i,k}$ for $i \in [m]$, $k \in [\ell]$ be a variable of $\mathrm{BPHP}_{2^\ell}^m$ indicating the ℓth bit of the ith string s_i. Let $Q_j(x_1, x_2, \ldots, x_\ell)$ for $j \in [2^k]$ be a multilinear polynomial over reals such that for all $a_1, a_2, \ldots, a_\ell \in \{0,1\}^\ell$, $Q_j(a_1, a_2, \ldots, a_\ell) = 1$ if $(a_1, a_2, \ldots, a_\ell) = \mathrm{bin}_\ell(j-1)$ and $Q_j(a_1, a_2, \ldots, a_\ell) = 0$ otherwise. We ma define Q_j as follows

$$Q_j(x_1, \ldots, x_\ell) = \prod_{k=1}^{\ell} (1 - x_k + \alpha_\ell)$$

for $i \in [m]$, $j \in [2^k]$, where $\alpha = \mathrm{bin}_\ell(j-1)$. By the construction $\deg(Q_j) = \ell$. Let $P_{i,j} = Q_j(s_{i,1}, s_{i,2}, \ldots, s_{i,\ell})$. Consider a tree-like $\mathrm{Th}(1)$-refutation of $\mathrm{PHP}_{2^\ell}^m$ of size S: $f_1 \geq 0, f_2 \geq 0, \ldots, f_S \geq 0$, where f_i are linear real polynomials over variables $p_{i,j}$ and $f_S \geq 0$ is unsatisfiable on Boolean cube. For each of the inequalities on the following conditions hold:

(a) $f_i \geq 0$ is semantically implied by $f_j \geq 0$ and $f_k \geq 0$ on the Boolean cube for $j, k < i$.

(b) f_i is a linear representation of an axiom of $\mathrm{PHP}_{2^\ell}^m$;

Let F_i be a polynomial obtained of substitution $p_{j,k} := P_{j,k}$ to f_i for all $j \in [m]$; $k \in [2^\ell]$. Consider a sequence of inequalities $F_1 \geq 0, \ldots, F_S \geq 0$. Observe that $F_S \geq 0$ is unsatisfiable on the Boolean cube since $P_{i,j} \in \{0,1\}$ on the Boolean cube. Let us verify that the sequence $F_1 \geq 0, \ldots, F_S \geq 0$ may be extended to a correct tree-like $\mathrm{Th}(\ell)$ refutation of $\mathrm{BPHP}_{2^\ell}^m$:

(a) If $f_i \geq 0$ is semantically implied by $f_j \geq 0$ and $f_k \geq 0$, then $F_i \geq 0$ is also implied by $F_j \geq 0$ and $F_k \geq 0$, since $P_{i,j}$ is Boolean on the Boolean cube.

(b) If f_i is a linear representation of a

- hole axiom then $f_i \geq 0$ is equivalent to the function $(1 - p_{a,b}) + (1 - p_{c,b}) \geq 1$ on $\{0,1\}^{\mathrm{Vars}(\mathrm{PHP}_{2^\ell}^m)}$ for $a, c \in [m]$, $b \in [2^\ell]$. Thus $F_i \geq 0$ is also equivalent to $(1 - P_{a,b}) + (1 - P_{c,b}) \geq 1$ on the Boolean cube. Observe that the restriction of $(1 - P_{a,b}) + (1 - P_{c,b}) \geq 1$ to the Boolean cube coincides with the predicate $s_a \neq \mathrm{bin}_\ell(b) \vee s_c \neq \mathrm{bin}_\ell(b)$ which is an axiom of $\mathrm{BPHP}_{2^\ell}^m$.
- pigeon axiom then $f_i \geq 0$ is equivalent to $\sum_{j=1}^{2^\ell} p_{a,j} \geq 1$ on the Boolean cube for some $a \in [m]$. Thus $F_i \geq 0$ is equivalent to $\sum_{j=1}^{2^\ell} P_{a,j} \geq 1$ on $\{0,1\}^{\mathrm{Vars}(\mathrm{BPHP}_{2^\ell}^m)}$. Observe that the latter inequality is identically true, since $P_{a,j}$ is equivalent to $s_a = \mathrm{bin}_\ell(j-1)$, so for exactly one value of $j \in [2^\ell]$, $P_{a,j} = 1$. Since $F_i \geq 0$ is identically true it can be semantically derived from two arbitrary axioms of $\mathrm{BPHP}_{2^\ell}^m$.

It is easy to see that the size of the resulting refutation is at most $3S$. □

Proposition44 ([4]). For $m > n$ there is a tree-like Cutting Planes (a subsystem of Th(1)) refutation of PHP_n^m of size $O(m^2 n)$.

Proposition6. For $m > 2^\ell$ there is a tree-like Th(ℓ) refutation of $\mathrm{BPHP}_{2^\ell}^m$ of size $O(m^2 2^\ell)$.

Proof. Follows from Propositions43 and44. □

A Proof of Lemma 12

Lemma 12. Let T be a binary tree with m vertices such that the ith vertex is labeled with $a_i \in \{0,1\}$ with the hereditary property: for each inner vertex i with direct descendants c_1 and c_2, if $a_i = 1$, then $a_{c_1} = 1$ or $a_{c_2} = 1$. We also assume that if r is the root of T, then $a_r = 1$. Assume that we have a one-sided bounded error oracle access to a_i i.e. if we request a value of a_i and $a_i = 0$ we get 1 with probability at most 1/2 and 0 with probability at least 1/2; if $a_i = 1$ we get 1 with probability 1. Then there exists an algorithm A that with probability at least 2/3 returns a leaf ℓ of T with $a_\ell = 1$ and makes $O(\log m)$ oracle queries to $a_1, \ldots, a_m$.

Proof of Lemma 12. For a tree F we denote by $|F|$ the number of nodes in F and for a node v of F we denote by $\mathrm{Subtree}(F, v)$ the subtree of F with root v. Let $\mathrm{Oracle}(i)$ be the oracle function returning the correct value of a_i with probability at least 9/10. We can implement such a function using the majority vote of a constant number of initial oracle queries. Let C be a constant; an appropriate value of C we choose later. Consider Algorithm 2 on the following page. We claim that at any iteration T_i has the hereditary property. This is the case in the beginning and if i decreases at some iteration, then the next T_i was considered at an earlier iteration. Otherwise, the next T_i is either a subtree of the current T_i (in that case the hereditary property is clearly maintained), or is obtained by removal a subtree with 0-labeled root (here we use that the oracle has a one-sided error) from the previous T_i (the hereditary property is also maintained in that case). We first consider a variant of the algorithm that works infinitely long (i.e., $C = +\infty$) and compute the expected number of the first iteration such that T_i consists of a single 1-labeled leaf of T. Notice that after the first such iteration the value of T_i stays the same for all further iterations. We show that that the expected value is at most $C \log m$ for some constant C. Then by running the algorithm for $3C\lceil \log m \rceil$ iterations we obtain the required error probability by Markov's inequality.

Algorithm 2: Search for 1-leaf.

$T_0 := T$ ▷ Initialize the tree
$i := 0$
for $j := 1$ to $3C\lceil \log_{3/2} m \rceil$ **do**
  $r :=$ root of T_i
  if $\mathrm{Oracle}(r) = 0$ **then**

$i := \max\{0, i-1\}$ ▷ Backtrack since the current tree may not contain a 1-leaf
else if $|T_i| \neq 1$ **then**
$v :=$ a centroid node of T_i
▷ i.e. such that $|\text{Subtree}(T_i, v)| \in [\frac{1}{3}|T_i|, \frac{2}{3}|T_i|]$
if Oracle$(v) = 1$ **then**
$T_{i+1} := \text{Subtree}(T_i, v)$
else
$T_{i+1} := T_i - \text{Subtree}(T_i, v)$
▷ T_{i+1} is obtained from T_i by the deletion of Subtree(T_i, v)
end if
$i := i + 1$
end if
end for
return the only node of T_i, if $|T_i| = 1$

Let $T(j)$ denote the value of T_i before the start of jth iteration, $i(j)$ denote i at the start of jth iteration and $r(j)$ denote the root of $T(j)$. Notice that if $a_{r(j)} = 1$, then for every $j' > j$, $T(j')$ is a subtree of $T(j)$, since the algorithm never backtracks if the true value of the roots label is 1. Hence, if $a_{r(j)} = a_{r(j')} = 1$ for some $j < j'$, then $i(j) \leq i(j')$. Let us consider a sequence $j_1, j_2, j_3, \ldots$, where $j_1 = 0$, $j_s = \min\{j \mid a_{r(j)} = 1 \wedge j > j_{s-1} \wedge i(j) > i(j_{s-1})\}$, if such minimum exists. Let us consider the iterations from j_s till $j_{s+1} - 1$. We consider the random variables $Y_{j_s}, Y_{j_s+1}, \ldots, Y_{j_{s+1}-1}$ corresponding to these iterations with the following properties:

- If $T(j)$ coincides with $T(j_s)$, then its root is labeled with 1. Then $Y_j = -1$ if the second oracle query returns the correct answer and $Y_j = 1$ if the answer it incorrect. Notice that $\Pr[Y_j = -1] \geq 9/10$.
- If the root of $T(j)$ is labeled with zero, then $Y_j = -1$, if the first oracle query returns the correct answer (i.e. the algorithm backtracks). Otherwise, if $T(j)$ consists of a single node $Y_j = 0$. Otherwise, if the root of $T(j+1)$ is labeled with 0, then $Y_j = 1$. If it is labeled with 1, then $Y_j = -\infty$. Notice that $\Pr[Y_j \leq -1] \geq 9/10$.

Notice that, $j_{s+1} = j_s + \min\{k \mid \sum_{j=j_s}^{j_s+k-1} Y_j \leq -1\}$. In order to estimate the expected value of $j_{s+1} - j_s$ we consider an auxiliary random variables $X_{j_s}, X_{j_s+1}, \ldots, X_{j_{s+1}-1}$, defined as

$$X_j = \begin{cases} 1, & \text{if } Y_j \geq 0 \\ -1, & \text{if } Y_j < 0. \end{cases}$$

Notice then $\sum_{j=j_s}^{j_s+k-1} Y_j \leq \sum_{j=j_s}^{j_s+k-1} X_j$. We can apply the following fact about random walks in a straight line to the random variables X_j:

Theorem 45 (Section XII.2 of [18]). Let $X_1, X_2, \ldots$ be a sequence of independent random variables that take value in $\{-1, 1\}$. Assume that for all i, $\Pr[X_i = 1] \leq 1/10$ and $\Pr[X_i = -1] \geq 9/10$. Let M be a random variable that equals the minimal natural number k such that $\sum_{i=1}^{k} X_i = -1$. Then the expected value of M is at most C, where $C \in \mathbb{R}$ is an absolute constant.

Fact 45 implies that $\mathbb{E}[j_{s+1} - j_s] \leq C$. Then $\mathbb{E}[j_s] = \mathbb{E}[j_s - j_{s-1} + (j_{s-1} - j_{s-2}) + \cdots + (j_2 - j_1) + (j_1 - j_0)] \leq sC$. Thus, by Markov's inequality $\Pr[j_s \leq 3sC] \geq 2/3$. Since $|T_{j_s}| \leq (2/3)^s |T_{j_0}|$, the algorithm that runs for $3C\lceil \log_{3/2} m \rceil$ iterations terminates in a 1-labeled leaf with probability at least $2/3$. □

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